

WEEK 13 — Unit 5: Continuous Integration (CI) with GitHub Actions

Topic: Workflow Automation, Events, Triggers, Key Components — Workflows, Jobs, Steps, Actions & Runners
Course: INT 332 – DevOps

Note 1: Understanding Workflow Automation & GitHub Actions Core Concepts

What is Workflow Automation?

In a software development team, many tasks happen repeatedly every time code changes — compiling the code, running tests, checking code style, building a Docker image, and deploying to a server. Doing all of these manually after every commit is slow, inconsistent, and error-prone. A developer might forget to run tests before pushing, or deployment steps might differ between team members.

Workflow automation means defining these repetitive tasks as code so they run automatically in response to events — a push to a branch, a pull request being opened, a scheduled time, or a manual trigger. This is the foundation of Continuous Integration (CI) and Continuous Delivery (CD).

What is GitHub Actions?

GitHub Actions is GitHub's built-in automation platform that allows you to define automated workflows directly inside your repository. Workflows are written as YAML files and stored in a special directory in your repository. When a defined event occurs — such as a developer pushing code — GitHub automatically runs the workflow on a virtual machine called a runner.

Why GitHub Actions Over Other CI Tools?

- Deeply integrated into GitHub — no separate server or account needed
- Free tier available for public repositories
- Workflows live in the same repository as the code — versioned together
- Massive marketplace of pre-built actions for common tasks
- Supports any language, framework, or platform
- Matrix builds allow testing across multiple versions simultaneously
- Secrets management is built in

The Workflow Directory Structure

GitHub Actions workflows must be placed in a specific directory inside your repository:

my-project/

└─ .github/

| └─ workflows/

| └─ ci.yml

| └─ release.yml

| └─ deploy.yml

└─ src/

└─ pom.xml

└─ Dockerfile

- The .github/ directory is at the root of the repository
- All workflow files go inside .github/workflows/
- Each workflow is a separate .yml file
- You can have multiple workflow files — each handles a different concern (CI, release, deployment, scheduled tasks)
- Workflow files are automatically detected and registered by GitHub — no additional configuration needed

Key Terminology

Workflow A workflow is the top-level automation unit. It is defined in a single YAML file and contains one or more jobs. A workflow is triggered by an event. A repository can have multiple workflows.

Event An event is what triggers a workflow to run. Events can be GitHub-specific actions (push, pull request, release) or external triggers (webhooks, schedules, manual dispatch).

Job A job is a set of steps that execute on the same runner (virtual machine). Jobs within a workflow run in parallel by default. You can configure dependencies between jobs to make them run sequentially.

Step A step is a single task within a job. Steps run sequentially within a job on the same runner. A step is either a shell command or a reference to a pre-built action.

Action An action is a reusable unit of code that performs a specific task — checking out code, setting up Java, logging into Docker Hub, sending a Slack notification. Actions are the

building blocks of steps. They can be official GitHub actions, community-published actions from the marketplace, or custom actions you write yourself.

Runner A runner is the virtual machine that executes the jobs in a workflow. GitHub provides hosted runners (Ubuntu, Windows, macOS). You can also set up self-hosted runners on your own infrastructure.
